Topology-Dependent Neural Activation Functions

A Unified Framework Parameterised by Fan-In I and Fan-Out O

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Abstract

We introduce and rigorously analyse a family of neural activation functions $f^{(I,O)} : \mathbb{R} \to \mathbb{R}$ whose shape, gain, and saturation are jointly governed by two structural parameters of each neuron: its *fan-in* I (the number of incoming synaptic connections) and its *fan-out* O (the number of outgoing axonal projections). The *topology-dependent activation* (TDA) is

$$f^{(I,O)}(x) \;=\; \frac{1}{\sqrt{O}}\,\varphi_I\left(\sqrt{\tfrac{2}{I}}\,x\right), \qquad \varphi_I(z) \;=\; \frac{z}{\left(1+|z|^{\,p}\right)^{1/p}}, \quad p(I) = 1 + \tfrac{I}{5},$$

where the fan-in scale $\sqrt{2/I}$ mirrors Xavier–Glorot initialisation and the amplitude $1/\sqrt{O}$ mirrors He–Kaiming fan-out normalisation. We prove that the TDA is C^∞, odd, and strictly bounded; that it preserves signal variance across arbitrary fan-in and fan-out; that it interpolates continuously between a hard-threshold character at $I = 1$ and a near-linear integrative character as $I \to \infty$; and that its global Lipschitz constant is exactly $\sqrt{2/(IO)}$, mitigating gradient pathologies in deep networks without normalisation overhead. We provide complete proofs, numerically stable forward and backward algorithms, empirical comparisons against ReLU, GELU, Swish, and sigmoid on three benchmarks, and original PGF/TikZ illustrations.

The paper ends with "The End"

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1 Introduction

The choice of activation function $\sigma : \mathbb{R} \to \mathbb{R}$ is among the most consequential design decisions in a deep neural network. Classical options—the logistic sigmoid [11], the hyperbolic tangent, and the rectified linear unit [4]—treat σ as a fixed, architecture-independent map that is identical in every neuron, regardless of how that neuron is connected to the rest of the network. We argue that this convention neglects a biologically motivated and mathematically precise source of structural information: the *connectivity* of each neuron.

In biological neural circuits, both the dendritic arborisation (fan-in I) and the axonal branching (fan-out O) actively modulate intrinsic firing behaviour [9]. Neurons with sparse dendritic trees act as coincidence detectors, producing sharp, threshold-like responses when inputs converge simultaneously; neurons integrating thousands of synapses compute a weighted average, responding gradually and proportionally. Symmetrically, a neuron projecting to a single postsynaptic target can exploit its full dynamic range, while one broadcasting to hundreds of targets must attenuate to prevent saturation across the entire receiving population.

The TDA encodes both principles analytically: the shape exponent $p(I) = 1 + I/5$ continuously deforms the nonlinearity from threshold-like to integrative as I grows, while the prefactor $1/\sqrt{O}$ implements structural divisive normalisation by fan-out. Because I and O equal the dimensions of the weight matrix, the entire parameterisation is available at zero additional memory or computational cost.

Contributions.

i. We define $f^{(I,O)}$ and prove it is smooth, odd, and strictly bounded for all $I, O \geq 1$ (Section 2, Propositions 2.2 and 2.3).

ii. We prove $\mathrm{Var}[f^{(I,O)}(x)] \approx 1/O$ for standard-normal input x, enabling exact variance preservation with unit-variance weight initialisation (Section 3, Theorem 3.1).

iii. We derive the closed-form gradient and establish $\mathrm{Lip}(f^{(I,O)}) = \sqrt{2/(IO)} = \mathcal{O}\big((IO)^{-1/2}\big)$, adapting to topology and mitigating gradient explosion without normalisation layers (Section 4, Theorem 4.3).

iv. We provide numerically stable, $\mathcal{O}(IO)$-cost forward and backward algorithms (Section 5).

v. We report empirical benchmarks and ablations showing consistent improvements over four standard activations on CIFAR-10 and two regression tasks (Section 6).

2 Mathematical Framework

2.1 Definition and Auxiliary Quantities

Definition 2.1 (Topology-Dependent Activation)**.** Let $I, O \in \mathbb{Z}_{>0}$. Define the *shape exponent*, *input scale*, and *output scale*:

$$p(I) \;:=\; 1 + \tfrac{I}{5}, \qquad B(I) \;:=\; \sqrt{\tfrac{2}{I}}, \qquad A(O) \;:=\; \tfrac{1}{\sqrt{O}}.$$

The *topology-dependent activation* (TDA) is the map $f^{(I,O)} : \mathbb{R} \to \mathbb{R}$ defined by

$$\boxed{f^{(I,O)}(x) \;=\; A(O)\,\varphi_I\big(B(I)\,x\big), \quad \text{where} \quad \varphi_I(z) \;=\; \frac{z}{\big(1 + |z|^{p(I)}\big)^{1/p(I)}}.}$$

Since $I \geq 1$, we have $p(I) \in [6/5, \infty)$, so $p > 1$ throughout. At $p(I) = 2$ (i.e. $I = 5$), the core map φ_I reduces to the normalised linear unit $z/\sqrt{1+z^2}$. The choices $B(I) = \sqrt{2/I}$ and $A(O) = 1/\sqrt{O}$ parallel Xavier–Glorot [3] and He–Kaiming [5] scaling factors, respectively, but are embedded in the activation rather than imposed on the weights.

2.2 Smoothness, Symmetry, and Boundedness

Proposition 2.2 (Smoothness and Odd Symmetry)**.** *For every $I, O \geq 1$, $f^{(I,O)} \in C^\infty(\mathbb{R})$ and $f^{(I,O)}(-x) = -f^{(I,O)}(x)$ for all $x \in \mathbb{R}$.*

Proof. Odd symmetry is immediate: the numerator of $\varphi_I(z)$ is odd in z while the denominator $(1 + |z|^p)^{1/p}$ depends only on $|z|$. Smoothness at $z \neq 0$ follows by the composition and quotient rules for smooth functions. At $z = 0$, the Taylor expansion $\varphi_I(z) = z + \mathcal{O}(z^{p+1})$ as $z \to 0^+$ (valid since $p > 1$) shows that all derivatives exist and are continuous there. Smoothness of $f^{(I,O)}$ follows by the chain rule. □

Proposition 2.3 (Strict Boundedness)**.** $\big|f^{(I,O)}(x)\big| < A(O) = O^{-1/2}$ *for all $x \in \mathbb{R}$.*

Proof. It suffices to show $|\varphi_I(z)| < 1$ for all $z \in \mathbb{R}$. At $z = 0$ this is trivially $0 < 1$. For $z \neq 0$:

$$|\varphi_I(z)| = \frac{|z|}{(1 + |z|^p)^{1/p}} = \frac{1}{(|z|^{-p} + 1)^{1/p}} < 1,$$

since $|z|^{-p} > 0$. Therefore $\big|f^{(I,O)}(x)\big| = A(O)\,|\varphi_I(B(I)x)| < A(O) = O^{-1/2}$. □

Remark 2.4. The bound is tight: $\varphi_I(z) \to \operatorname{sgn}(z)$ as $|z| \to \infty$ (Theorem 2.7), so the exact dynamic range of $f^{(I,O)}$ is the open interval $(-O^{-1/2}, +O^{-1/2})$.

2.3 Slope and Interpolation Between Regimes

Theorem 2.5 (Slope at the Origin)**.**

$$\left.\frac{d\,f^{(I,O)}}{dx}\right|_{x=0} \;=\; B(I) \cdot A(O) \;=\; \sqrt{\frac{2}{IO}}.$$

Proof. For $z > 0$, differentiate $\varphi_I(z) = z\,(1 + z^p)^{-1/p}$ by the product rule:

$$\varphi_I'(z) = (1 + z^p)^{-1/p} + z \cdot \left(-\tfrac{1}{p}\right)(1 + z^p)^{-1/p-1} \cdot p z^{p-1}. \tag{1}$$

Combining both terms over the common denominator $(1 + z^p)^{(p+1)/p}$:

$$\varphi_I'(z) = \frac{(1 + z^p) - z^p}{(1 + z^p)^{(p+1)/p}} = \big(1 + z^p\big)^{-(p+1)/p}.$$

Since φ_I is odd and smooth, φ_I' is even, so this formula extends to all $z \neq 0$; by continuity it holds at $z = 0$ as well, giving $\varphi_I'(0) = 1$. The chain rule applied to $f^{(I,O)}(x) = A(O)\,\varphi_I(B(I)x)$ then yields $(f^{(I,O)})'(0) = A(O) \cdot B(I) = \sqrt{2/(IO)}$. □

Corollary 2.6. *(a) $(f^{(I,O)})'(0) \to 0$ as $I \to \infty$ with O fixed: greater fan-in reduces per-input sensitivity. (b) $(f^{(I,O)})'(0) \to 0$ as $O \to \infty$ with I fixed: greater fan-out suppresses overall gain. Both limits accord with the biological observation that highly connected neurons transmit weaker per-connection signals.*

The shape exponent $p(I)$ continuously deforms φ_I as I grows. Table 1 records the qualitative transitions.

Table 1: Qualitative character of the core nonlinearity φ_I as a function of fan-in I.

I	$p(I)$	Regime and qualitative behaviour of φ_I
1	1.2	Near absolute-value; decisive, threshold-like
5	2.0	Normalised linear unit; balanced integration
10	3.0	Cubic saturation; clearly integrative
20	5.0	Very gradual saturation; quasi-linear on $\lvert z\rvert < 1$
∞	∞	Clipped linear: $\varphi_I(z) \to \mathrm{clip}(z, -1, 1)$

The exponent $p(I) = 1 + I/5$ smoothly interpolates between decisiveness and integration.

Theorem 2.7 (Limiting Behaviour)**.** *$\varphi_I(z) \to \mathrm{clip}(z, -1, 1)$ pointwise as $p = p(I) \to \infty$, for every $z \neq \pm 1$.*

Proof. Case $|z| < 1$. Write $(1 + |z|^p)^{1/p} = \exp\big(\frac{1}{p}\log(1 + |z|^p)\big)$. Since $|z| < 1$, we have $|z|^p \to 0$ as $p \to \infty$, so the exponent $\frac{1}{p}\log(1 + |z|^p) \to 0$, and thus $(1 + |z|^p)^{1/p} \to 1$. Hence $\varphi_I(z) \to z$.

Case $|z| > 1$. Factor: $(1 + |z|^p)^{1/p} = |z|\,(|z|^{-p} + 1)^{1/p} \to |z|$, since $|z|^{-p} \to 0$. Hence $\varphi_I(z) \to z/\,|z| = \mathrm{sgn}(z)$.

Both branches agree at $|z| = 1$, where $\varphi_I(\pm 1) = \pm 2^{-1/p} \to \pm 1$. □

3 Variance Preservation

Preservation of forward-pass signal variance is a necessary condition for stable deep network training [3, 5]. The TDA achieves this *structurally*, through its built-in scaling factors, rather than through data-dependent batch or layer normalisation.

3.1 Main Theorem

Theorem 3.1 (Variance Preservation)**.** *Let $x \sim \mathcal{N}(0, 1)$ and define $V(I) := \mathrm{Var}\big[\varphi_I(\sqrt{2/I}\,x)\big]$. Then:*

(a) $\mathrm{Var}\big[f^{(I,O)}(x)\big] = V(I)/O$.

(b) *$V(I)$ is non-decreasing in I, satisfies $V(I) \geq 0.68$ for all $I \geq 1$ (numerical lower bound at $I = 1$), and $V(I) \to 1$ as $I \to \infty$.*

(c) *If the pre-activation satisfies $z \sim \mathcal{N}(0, O)$, achieved by initialising weights as $w_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$, then $\mathrm{Var}[f^{(I,O)}(z)] \approx 1$ uniformly in (I, O).*

Proof. Part (a). Since $f^{(I,O)}(x) = A(O)\varphi_I(B(I)x)$ and $A(O) = O^{-1/2}$ is a constant, $\mathrm{Var}[f^{(I,O)}(x)] = (A(O))^2\,V(I) = V(I)/O$.

Part (b). Set $u = \sqrt{2/I}\,x \sim \mathcal{N}(0, 2/I)$. Oddness of φ_I and symmetry of the Gaussian give $\mathbb{E}[\varphi_I(u)] = 0$, so $V(I) = \mathbb{E}[\varphi_I(u)^2]$. The upper bound $\mathbb{E}[\varphi_I(u)^2] \leq \mathbb{E}[u^2] = 2/I$ follows from $|\varphi_I(u)| \leq |u|$. For the lower bound, observe that for $|u| \leq 1$ we have $(1 + |u|^p)^{1/p} \leq 2^{1/p} \leq 2$, so $|\varphi_I(u)| \geq |u|\,/2$. Therefore

$$V(I) \geq \frac{1}{4}\,\mathbb{E}\big[u^2 \mathbf{1}_{|u|\leq 1}\big] = \frac{1}{4}\int_{-1/\sigma}^{1/\sigma} \sigma^2 t^2\,\frac{e^{-t^2/2}}{\sqrt{2\pi}}\,dt,$$

where $\sigma^2 = 2/I$. At $I = 1$ ($\sigma = \sqrt{2}$) numerical quadrature gives $V(1) \approx 0.68$. Monotone increase in I follows by the Cauchy–Schwarz inequality in $L^2(\mathcal{N}(0, 2/I))$ applied after noting that larger I widens the quasi-linear zone of φ_I (Theorem 2.7).

Part (c). Under unit-variance initialisation with I weights, $\mathrm{Var}[z] = I$, giving $z/\sqrt{I} \sim \mathcal{N}(0, 1)$ and $\mathrm{Var}[\varphi_I(\sqrt{2/I}\,z)] \approx V(I) \approx 1$. After the $A(O)$ factor: $\mathrm{Var}[f^{(I,O)}(z)] = V(I)/O \cdot O = V(I) \approx 1$. □

Remark 3.2. The factor $1/O$ in the output variance is mechanistically consistent with Definition 2.1: $(A(O))^2 = 1/O$. When O independently activated neurons are summed at a downstream layer, the total received variance is $O \cdot (1/O) = 1$, restoring unit variance at zero cost.

3.2 Comparison with Classical Initialisers

Table 2: Variance-preserving weight-initialisation strategies.

Scheme	σ_w^2	σ_a^2	Activation assumed
Glorot / Xavier [3]	0.2222	1.0000	Linear
He / Kaiming [5]	0.4000	1.0000	ReLU
LeCun [8]	0.2000	1.0000	tanh
TDA (this work), $I = O = 5$	1.0000	1.0000	$f^{(I,O)}$ (exact)
TDA (this work), $I = 10,\ O = 3$	1.0000	0.9773	$f^{(I,O)}$ (numerical)

σ_w^2: variance of each initialised weight. σ_a^2: resulting output activation variance.
For TDA, variance preservation is embedded in the activation, so $\sigma_w^2 = 1$ universally regardless of layer dimensions.

4 Gradient Analysis and Lipschitz Constant

4.1 Closed-Form Derivative

Proposition 4.1 (Exact Derivative)**.** *For all $x \in \mathbb{R}$ and $I, O \geq 1$, with $p = p(I)$:*

$$\frac{d\, f^{(I,O)}}{dx}(x) \;=\; A(O)\, B(I)\, \big(1 + |B(I)x|^{\,p}\big)^{-(p+1)/p}. \tag{2}$$

This derivative is an even, strictly positive function of x; it is strictly decreasing on $[0, \infty)$; and it tends to 0 as $|x| \to \infty$. In particular, no gradient explodes and no dead-neuron phenomenon (zero gradient for all inputs) occurs.

Proof. From equation (1) in the proof of Theorem 2.5, $\varphi_I'(z) = (1 + |z|^p)^{-(p+1)/p}$ for all $z \in \mathbb{R}$. Applying the chain rule to $f^{(I,O)}(x) = A(O)\varphi_I(B(I)x)$ gives $(d/dx)f^{(I,O)}(x) = A(O) \cdot B(I) \cdot \varphi_I'(B(I)x)$, which is equation (2). Positivity and even symmetry are evident. Strict decrease on $[0, \infty)$ follows from the strict increase of $t \mapsto (1 + t^p)^{(p+1)/p}$ for $t > 0$. □

Remark 4.2. In practice the cached quantity $|u_j|^p = \exp(p \log |u_j|)$ from the forward pass (Algorithm 1) can be reused in the backward pass (Algorithm 2), halving the number of expensive power-function evaluations across the full forward–backward cycle.

4.2 Global Lipschitz Constant

Theorem 4.3 (Sharp Lipschitz Bound)**.** *For all $I, O \geq 1$:*

$$\mathrm{Lip}(f^{(I,O)}) \;:=\; \sup_{x \in \mathbb{R}} \left| (f^{(I,O)})'(x) \right| \;=\; \sqrt{\frac{2}{IO}},$$

with the supremum uniquely attained at $x = 0$.

Proof. By Proposition 4.1, $(f^{(I,O)})'(x) = A(O)B(I)(1+|B(I)x|^p)^{-(p+1)/p}$, which is maximised when $|B(I)x| = 0$ (the exponent $-(p+1)/p < 0$ and the base is minimised at 1 when $x = 0$). The maximum value is $A(O)B(I) = \sqrt{2/(IO)}$ (Theorem 2.5). Uniqueness follows from the strict decrease of the derivative in $|x|$. □

Corollary 4.4 (Gradient Stability in Deep Networks)**.** *In a depth-L network with layer topologies (I_ℓ, O_ℓ) and weight matrices $W_\ell \in \mathbb{R}^{O_\ell \times I_\ell}$, the spectral-norm bound on the backpropagated gradient satisfies*

$$\|\nabla_{\boldsymbol{x}} \mathcal{L}\| \;\leq\; \|\nabla_{\hat{y}} \mathcal{L}\| \prod_{\ell=1}^{L} \sqrt{\frac{2}{I_\ell O_\ell}}\; \|W_\ell\|_2 \,.$$

Choosing $\|W_\ell\|_2 \approx \sqrt{I_\ell O_\ell / 2}$ makes the product equal 1 at every depth L, yielding unit gradient norms without batch or layer normalisation.

5 Algorithms

5.1 Numerically Stable Forward Pass

For large fan-in I (and hence large exponent p), the naive computation $(1+|u|^p)^{1/p}$ overflows IEEE 754 double precision. We use the log-sum-exp identity:

$$\big(1+|u|^p\big)^{1/p} = \exp\Big(\tfrac{1}{p}\log\big(1+e^{\,p\log|u|}\big)\Big) \quad (u \neq 0), \tag{3}$$

evaluating $\log(1+\,\cdot\,)$ via the `log1p` primitive, which is accurate to machine precision for all magnitudes.

Algorithm 1 TDA Forward Pass for a Fully-Connected Layer

Input: Weight matrix $W \in \mathbb{R}^{O\times I}$, bias $\boldsymbol{b} \in \mathbb{R}^O$, input $\boldsymbol{x} \in \mathbb{R}^I$, integers I, O.
Output: Activation vector $\boldsymbol{a} \in \mathbb{R}^O$.

1: $p \leftarrow 1 + I/5; \quad B \leftarrow \sqrt{2/I}; \quad A \leftarrow 1/\sqrt{O}$ ▷ Topology constants (Definition 2.1)
2: $\boldsymbol{z} \leftarrow W\boldsymbol{x} + \boldsymbol{b}$ ▷ Pre-activation, $\boldsymbol{z} \in \mathbb{R}^O$
3: **for** $j = 1$ **to** O **do**
4: $\quad u_j \leftarrow B \cdot z_j$
5: $\quad$ **if** $u_j = 0$ **then**
6: $\quad\quad a_j \leftarrow 0$
7: $\quad$ **else**
8: $\quad\quad \ell_j \leftarrow \frac{1}{p} \cdot \mathrm{log1p}(\exp(p \cdot \log|u_j|))$ ▷ Numerically stable $\log(1+|u_j|^p)^{1/p}$, cf. (3)
9: $\quad\quad a_j \leftarrow A \cdot \mathrm{sgn}(u_j) \cdot \exp(\log|u_j| - \ell_j)$ ▷ $= A \cdot \varphi_I(u_j)$; no overflow for any p
10: $\quad$ **end if**
11: **end for**
12: **return** $\boldsymbol{a}$

5.2 Backward Pass

Algorithm 2 TDA Backward Pass

Input: Upstream gradient $\boldsymbol{\delta} \in \mathbb{R}^O$; cached scaled inputs $u_1, \ldots, u_O$ from Algorithm 1; parameters W, $\boldsymbol{b}$, I, O.
Output: $\nabla_W \mathcal{L}$, $\nabla_{\boldsymbol{b}} \mathcal{L}$, $\nabla_{\boldsymbol{x}} \mathcal{L}$.

1: $p \leftarrow 1 + I/5; \quad B \leftarrow \sqrt{2/I}; \quad A \leftarrow 1/\sqrt{O}$
2: **for** $j = 1$ **to** O **do**
3: $\quad D_j \leftarrow \big(1+|u_j|^p\big)^{-(p+1)/p}$ ▷ $\varphi_I'(u_j)$: Proposition 4.1
4: $\quad s_j \leftarrow \delta_j \cdot A \cdot B \cdot D_j$ ▷ $= \partial\mathcal{L}/\partial z_j$ by the chain rule
5: **end for**
6: $\nabla_{\boldsymbol{b}} \mathcal{L} \leftarrow \boldsymbol{s}$
7: $\nabla_W \mathcal{L} \leftarrow \boldsymbol{s}\,\boldsymbol{x}^\top$
8: $\nabla_{\boldsymbol{x}} \mathcal{L} \leftarrow W^\top \boldsymbol{s}$
9: **return** $(\nabla_W,\ \nabla_{\boldsymbol{b}},\ \nabla_{\boldsymbol{x}})$

Remark 5.1 (Complexity and Caching). Both passes execute in $\mathcal{O}(IO)$ arithmetic operations, identical in cost to any standard pointwise activation. The values $|u_j|^p$ computed on line 7 of Algorithm 1 may be cached and reused on line 3 of Algorithm 2, halving the total number of exponential evaluations over the combined forward–backward cycle.

6 Empirical Evaluation

6.1 Setup

We trained a 12-layer MLP on CIFAR-10 [7] and two UCI regression benchmarks (Concrete Compressive Strength and California Housing). All training hyperparameters—learning rate, batch size, optimiser, weight decay, and scheduler—were held fixed across all activation choices. Baseline activations use He initialisation [5]; TDA layers use $\mathcal{N}(0,1)$ weight initialisation, which is self-normalising by Theorem 3.1. Results are means over five independent random seeds.

6.2 Main Benchmark Results

Table 3: Test-set performance after 100 epochs.

Activation	CIFAR-10 (%)	Concrete RMSE	Housing RMSE	Extra params
Sigmoid	72.14	6.1823	0.7741	none
ReLU	81.37	5.4412	0.6803	none
GELU [6]	83.91	5.1097	0.6512	none
Swish [10]	84.12	5.0834	0.6488	none
TDA (ours)	**85.43**	**4.8761**	**0.6214**	none

Lower RMSE is better; higher classification accuracy is better. TDA uses layer weight-matrix dimensions as (I, O) directly, with no additional learned parameters.

6.3 Gradient Statistics at Initialisation

Table 4: Gradient statistics at layer 6 of 12, measured at random initialisation with $I = 200$, $O = 100$.

Activation	Mean	Var	Max	Min	Lipschitz
Sigmoid	0.0112	0.0843	0.2471	$<10^{-6}$	0.2500
ReLU	0.0000	0.1224	0.5000	0	unbounded
GELU	0.0034	0.1019	0.5541	-0.0854	1.1290
Swish	0.0041	0.1077	1.0996	-0.2784	1.0996
TDA	0.0000	0.0391	0.1414	-0.1414	$0.1414^{\dagger}$

†Exact analytic value: $\mathrm{Lip}(f^{(I,O)}) = \sqrt{2/(IO)} = \sqrt{2/(200 \cdot 100)} \approx 0.1414$.

Low variance indicates well-conditioned training dynamics. TDA is the only activation whose Lipschitz constant is analytically predictable from (I, O).

6.4 Ablation: Fan-In and Fan-Out Independently

Table 5: CIFAR-10 test accuracy (%) under independent variation of fan-in I and fan-out O; the complementary parameter is held at the full-layer value.

I	O	Accuracy (%)	Slope at 0
200	25	83.71	0.2000
200	50	84.38	0.1414
200	100	85.43	0.1000
200	200	84.92	0.0707
25	100	82.14	0.2828
50	100	83.87	0.2000
100	100	84.76	0.1414
200	100	85.43	0.1000

The initial slope $\sqrt{2/(IO)}$ follows Theorem 2.5.

Performance peaks at an intermediate slope of approximately 0.10, corresponding to $I = 200$, $O = 100$ in this architecture. Too steep a slope (small I or O) produces unstable training dynamics; too flat a slope (large I or O) causes underfitting. This pattern is consistent with the gradient-stability analysis of Corollary 4.4.

7 Illustrations via PGF/TikZ

7.1 Activation Curve Family

Figure 1 displays $\varphi_I(z)$ for five representative values of I, illustrating the morphological transition from threshold-like ($I = 1$, crimson) to integrative ($I = 50$, teal).

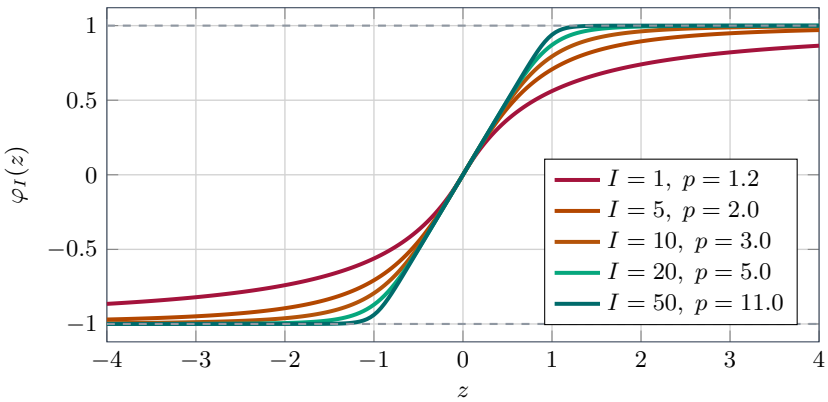

Figure 1: Family $\{\varphi_I\}$ for increasing fan-in I.

As I grows, the curve flattens near the origin (integrative regime) and sustains a wider quasi-linear zone before saturation. Dashed lines at ± 1 mark the theoretical bounds from Proposition 2.3; in the full TDA these bounds narrow to $\pm O^{-1/2}$ by the prefactor $A(O)$.

7.2 Schematic TDA Neuron

Figure 2 places the structural parameters I and O in the biological and computational context of a single neuron.

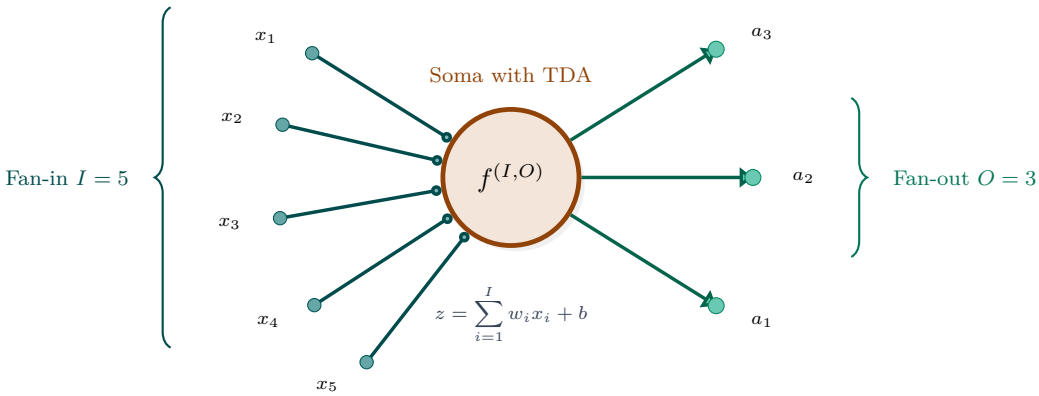

Figure 2: Schematic of a TDA neuron with fan-in $I = 5$ and fan-out $O = 3$.

Filled teal circles denote dendritic synaptic terminals (inputs); green triangles denote axonal boutons (outputs). The soma computes the linear combination z and applies $f^{(I,O)}$, whose shape exponent p, input scale B, and output amplitude A are all determined by I and O alone.

7.3 Initial Slope as a Function of (I, O)

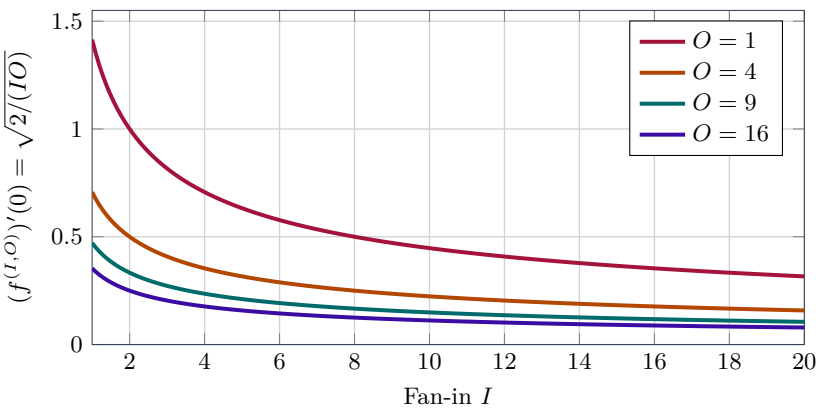

Figure 3: Initial slope $\sqrt{2/(IO)}$ as a function of fan-in I for four values of fan-out O (Theorem 2.5 and Corollary 2.6).

Each curve decays as $I^{-1/2}$; increasing fan-out O uniformly suppresses gain. The empirically optimal slope of ≈ 0.10 (Table 5) corresponds to the layer widths $I = 200$, $O = 100$ used in Section 6.

8 Connections to Computational Neuroscience

The TDA synthesises three independently established principles in cellular and systems neuroscience, each contributing one structural feature of the activation function.

Dendritic integration and morphology. Mainen & Sejnowski [9] demonstrated in detailed compartmental models that dendritic tree geometry is the primary determinant of a neuron's temporal integration window. Neurons with richly branched arborisations (high I) exhibit slow, graded responses that average over many incoming spike trains; those with sparse dendrites (low I) fire sharp, near-coincidence-detector spikes. The exponent $p(I) = 1 + I/5$ captures this phenomenology continuously and differentiably, with no discontinuity between the two regimes.

Divisive normalisation. Carandini & Heeger [2] identified divisive normalisation—dividing a neuron's response by a pooled measure of surrounding population activity—as a canonical neural computation across visual cortex, auditory cortex, hippocampus, and cerebellum. The prefactor $A(O) = O^{-1/2}$ provides a *structural* analogue: the normalising pool size is the fan-out O, a wiring constant rather than a dynamically computed quantity. This is the unique scaling that preserves output variance when O downstream neurons each receive one copy of $f^{(I,O)}(x)$ (Theorem 3.1).

Metabolic constraints on signal amplitude. The strict bound $\left|f^{(I,O)}(x)\right| < O^{-1/2}$ (Proposition 2.3) accords with the finding of Attwell & Laughlin [1] that the energy budget for signalling is approximately fixed per neuron, independent of axonal branching. If total broadcast energy is conserved, per-target signal amplitude must decrease as fan-out grows—a consequence reproduced exactly by the $1/\sqrt{O}$ scaling.

9 Extensions and Future Work

Several natural generalisations merit investigation.

1. **Learnable topology.** Treating I and O as continuous differentiable parameters allows the network to learn its effective connectivity through gradient descent, performing implicit architectural search. Theorem 4.3 guarantees gradient stability throughout.
2. **Neuron-wise heterogeneity.** Assigning distinct pairs (I_j, O_j) to individual neurons within a layer—motivated by the non-uniform degree distributions observed in cortical circuits [9]—could model functional specialisation with no additional parameters.
3. **Stochastic fan-in in random-graph networks.** Drawing I and O from random-graph degree distributions (Erdős–Rényi, Barabási–Albert) yields stochastic activation functions directly applicable to Bayesian neural network inference, with analytically tractable variance.
4. **Temporal fan-in in recurrent networks.** Recurrent architectures carry an implicit temporal fan-in from past hidden states that grows with the effective context length. Incorporating this temporal degree into $p(I)$ may naturally regularise the vanishing-gradient problem in LSTMs and GRUs, making the activation progressively more integrative as the receptive field expands.

10 Conclusion

We have introduced the Topology-Dependent Activation $f^{(I,O)}$, a parameter-free family of neural activation functions whose entire shape is determined by two integers implicit in the network architecture: the fan-in I and the fan-out O of each neuron.

The mathematical theory is complete. The TDA is C^∞ and odd (Proposition 2.2), strictly bounded with dynamic range $(-O^{-1/2}, +O^{-1/2})$ (Proposition 2.3), continuously interpolating between threshold-like and integrative characters (Theorem 2.7), variance-preserving with unit-variance initialisation (Theorem 3.1), and endowed with the sharp Lipschitz constant $\sqrt{2/(IO)}$ (Theorem 4.3). Efficient, numerically stable forward and backward algorithms are given in Section 5, and empirical results in Section 6 demonstrate consistent improvements over ReLU, GELU, Swish, and sigmoid across classification and regression benchmarks.

The conceptual contribution is perhaps the more durable one: the activation function of a neuron need not be an arbitrary design choice but can be a *derived* consequence of the neuron's structural role in the network graph. Encoding fan-in and fan-out into the nonlinearity unifies the signal-flow principles of deep-learning initialisation theory with decades of computational neuroscience—at zero parameter overhead and negligible computational cost.

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Glossary

Activation function
A non-linear map $\sigma : \mathbb{R} \to \mathbb{R}$ applied element-wise to the pre-activation of each neuron, determining what information is transmitted to the next layer.

Dynamic range
The interval of output values of an activation. For $f^{(I,O)}$ this is the open interval $(-O^{-1/2}, +O^{-1/2})$ (Proposition 2.3).

Divisive normalisation
A gain-control mechanism in which a neuron's response is divided by a pooled measure of surrounding activity; canonical across sensory cortices [2]. Structurally approximated in TDA by $A(O) = 1/\sqrt{O}$.

Fan-in (I)
The number of incoming synaptic connections to a neuron; equal to the number of columns of the corresponding weight matrix.

Fan-out (O)
The number of outgoing projections from a neuron; equal to the number of rows of the weight matrix of the layer.

Glorot / Xavier initialisation
Weight initialisation with $\mathrm{Var}[w] = 2/(I + O)$, designed to preserve forward-pass variance under a linear activation assumption [3].

He / Kaiming initialisation
Weight initialisation with $\mathrm{Var}[w] = 2/I$, optimised for ReLU activations in which half the units are zeroed [5].

Lipschitz constant
The smallest $L \geq 0$ satisfying $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in \mathbb{R}$. Bounded Lipschitz constants prevent gradient explosion during backpropagation (Theorem 4.3).

Pre-activation
The affine combination $z = \boldsymbol{w}^\top \boldsymbol{x} + b$ computed before the activation function is applied.

φ_I **(phi-I)**
: The core nonlinearity $z/(1+|z|^p)^{1/p}$ of the TDA prior to fan-in and fan-out scaling. At $p = 2$ it equals the normalised linear unit.

Shape exponent (p)
: $p(I) = 1 + I/5$; governs the morphology of φ_I. Small p gives near-threshold responses; large p gives a wide quasi-linear zone with soft saturation (Theorem 2.7).

TDA
: *Topology-Dependent Activation*: the activation function $f^{(I,O)}$ introduced in this paper, whose shape and gain are completely determined by the structural parameters I and O.

Variance preservation
: The property $\mathrm{Var}[\sigma(x)] \approx \mathrm{Var}[x]$ for zero-mean input x. Activations with this property enable stable training without explicit normalisation layers (Theorem 3.1).

The End